

DS-GA 1011: Natural Language Processing

Final Project Proposal (Track 1)

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1 Introduction

1.1 RAG: From Foundation to Adaptation

Retrieval-Augmented Generation (RAG) addresses fundamental limitations of Large Language Models (LLMs) by integrating external knowledge retrieval with text generation (Lewis et al., 2020). The Vanilla RAG architecture combines a neural retriever with a generator, treating retrieved documents as latent variables that condition generation. While this approach demonstrated significant improvements on open-domain question answering, it revealed critical limitations: indiscriminate retrieval regardless of necessity, inability to filter irrelevant passages, and lack of explicit training for document-grounded generation (Asai et al., 2023; Zhang et al., 2024). These shortcomings motivated two distinct research directions—adaptive retrieval with self-reflection (Self-RAG) and domain-specific fine-tuning with distractor training (RAFT)—each proposing fundamentally different solutions to improve how models leverage retrieved knowledge.

1.2 Comparing RAG Training Strategies for Multi-Hop Reasoning

Despite the proliferation of RAG variants, there remains limited empirical understanding of how different training strategies affect performance on complex reasoning tasks. While Self-RAG and RAFT both claim substantial improvements over Vanilla RAG, they represent fundamentally different design philosophies: Self-RAG introduces inference-time controllability through reflection tokens (Asai et al., 2023), whereas RAFT explicitly trains on distractor documents during fine-tuning (Zhang et al., 2024). Understanding which approach is more effective—and under what conditions—is critical for practitioners balancing training complexity, inference cost, and task performance. This project provides the first controlled comparison of three strategies—**adaptive retrieval** (Self-RAG), **robust filtering** (RAFT), and **passive augmentation** (Vanilla RAG)—using identical models (LLaMA 3.1-8B) and benchmarks (HotpotQA). By isolating the impact of retrieval and training mechanisms, we answer key questions: Does adaptive retrieval outperform explicit distractor training? How does each approach handle irrelevant passages? What are the computational trade-offs?

2 Project Proposal

We empirically test performance differences among three RAG systems: **Vanilla RAG** (Lewis et al., 2020), **Self-RAG** (Asai et al., 2023), and **RAFT** (Zhang et al., 2024). We control for both the backbone LLM and evaluation dataset, ensuring that observed differences arise from retrieval and training mechanisms rather than model or task variance. We use **LLaMA 3.1-8B** (Dubey et al., 2024) as the backbone model across all three systems, maintaining compatibility with the original papers while enabling fair comparison. This controlled setup allows us to isolate the impact of each method’s core innovation on multi-hop question answering.

2.1 RAG Systems

2.1.1 Vanilla RAG

Vanilla RAG establishes the foundational architecture for retrieval-augmented generation, combining parametric memory with non-parametric memory in a simple framework that can be trained end-to-end (Lewis et al., 2020). While it achieved state-of-the-art results on open-domain QA benchmarks and improved factuality compared to parametric-only models, it exhibits several limitations: retrieval quality directly constrains generation quality with no mechanism to handle poor retrieval, the model can experience “retrieval collapse” where it learns to ignore retrieved documents, and it still generates hallucinations when the knowledge base lacks necessary information. These limitations motivate the development of more sophisticated RAG approaches.

2.1.2 Self-RAG

Traditional RAG systems retrieve a fixed number of documents regardless of context, potentially introducing irrelevant information (Asai et al., 2023). Self-RAG addresses this by training models to retrieve adaptively through self-reflection mechanisms. The model generates a *Retrieve* token to determine if retrieval is beneficial, then uses reflection tokens to assess passage relevance (*ISREL*), evidence support (*ISSUP*), and output usefulness (*ISUSE*). This enables inference-time controllability via weighted reflection token probabilities. Self-RAG demonstrated significant gains in factuality and citation accuracy, outperforming larger proprietary models (Asai et al., 2023). However, Self-RAG introduces training complexity through its critic model and requires token-level probabilities at inference, limiting applicability to open-source models. For our project, it represents a middle ground between vanilla RAG and domain-specific training approaches.

2.1.3 RAFT

RAFT (Retrieval-Augmented Fine-Tuning) addresses Vanilla RAG’s limitations by explicitly teaching models how to use retrieved documents effectively (Zhang et al., 2024). Rather than passively providing context, RAFT fine-tunes the backbone LLM on training examples that include both relevant oracle documents and irrelevant distractor passages. This trains the model to identify useful documents and remain robust to noise, leading to more faithful grounding and lower hallucination rates. A potential limitation is that RAFT’s gains can be sensitive to domain alignment between training and target settings.

2.2 Evaluation Dataset

We selected HotpotQA (Yang et al., 2018) as our evaluation dataset because its multi-hop reasoning requirements directly test the core differentiators between RAG approaches: retrieval quality, distractor filtering, and information synthesis across documents. We evaluate using **Exact Match (EM)** and **F1 score**, which measure perfect answer matching and token-level overlap respectively. By focusing on this single, well-established dataset, we can conduct deeper analysis through extensive ablations and qualitative case studies, providing actionable insights into when and why each RAG training strategy succeeds or fails.

2.3 Project Milestones

Timeline & Phase	Tasks & Deliverables
Weeks 1-2 (Oct 15-25): Infrastructure and Dataset Setup	Download and preprocess HotpotQA, set up LLaMA-3.1 8B API, implement evaluation pipeline, and build Vanilla RAG baseline. Deliverable: Working Vanilla RAG system with EM/F1 evaluation confirmed.
Weeks 3-4 (Oct 26-Nov 7): Self-RAG Integration and Analysis	Configure Self-RAG 7B, implement reflection token inference, run threshold/weight experiments, and compare to Vanilla RAG. Deliverable: Self-RAG results with ablations and reflection token accuracy analysis.
Weeks 5-6 (Nov 8-20): RAFT Training and Evaluation	Prepare HotpotQA-based RAFT training data, fine-tune LLaMA-3.1 8B, and test different P% configurations. Deliverable: Trained RAFT model with results and ablation analysis.
Weeks 7-8 (Nov 21-Dec 3): Comprehensive Analysis and Writeup	Perform cross-model error/failure analysis, reasoning chain comparisons, efficiency benchmarks, manual evaluation, and statistical testing. Deliverable: Complete comparative analysis with figures, tables, and final report.

Contingency Plans: If training is slow, start with 50k instances instead of 150k. If GPU memory is limited, reduce batch size with gradient accumulation. Allocate 2-3 day buffer in training phase for debugging complex reflection token mechanism.

References

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